

Ocean Events

Eddies, fronts and upwelling detected from real satellite fields - with a clear line drawn between what is detected and what is merely encouraging.

PORT	PAGE FILE	MODULES	MODE
8506	app/phase2/events_page.py	phase2/events/{eddy, fronts, upwelling}.py	detection, not tracking

The ocean is not a smooth field. It is full of spinning eddies, sharp fronts and rising cold water, and these features are where fisheries concentrate, where cyclones gain or lose energy, and where the model is least certain. This page finds them.

1. What this feature is, in plain words

An **eddy** is a rotating body of water, often a hundred kilometres or more across, that can persist for months and carries its own heat and nutrients with it. Detection uses the Okubo-Weiss criterion, a standard method that separates regions dominated by rotation from regions dominated by stretching.

A **front** is a sharp horizontal temperature boundary - the ocean's equivalent of a weather front. Fish gather along them, and so do fishing fleets.

Upwelling is cold, nutrient-rich water rising from below, usually driven by wind pushing surface water away from a coast. It is the engine of some of the world's most productive fisheries, including along the Somali and Indian coasts.

Everything on this page is computed from genuine surface observations: satellite currents and satellite sea-surface temperature. You can switch to the reanalysis of the same days to see how closely the two agree.

THE EVERYDAY PICTURE

Weather radar for the sea surface. It does not tell you the temperature everywhere - it tells you where something is happening.

2. What you actually see on the screen

Three detectors, one date, one source toggle:

- An eddy map with each detected feature's centre, rotation sense and approximate diameter.
- A front map showing the strongest horizontal temperature gradients, with the detection threshold and the field's mean gradient both printed - so a flat field is visible as flat.
- An upwelling panel combining the surface signature with Ekman pumping from wind.
- A source toggle - **satellite** (genuine observations, the deliverable's own inputs) or **glorys** (the reanalysis of the same days).
- A clear on-screen statement that the upwelling panel's subsurface term uses reanalysis in both modes, and is therefore a signature rather than an attribution.

3. How to run it, step by step

RUN THIS

```
PYTHONPATH=src python -m streamlit run app/phase2/events_page.py --server.port 8506
```

- 1 Run the command and open **http://localhost:8506**.
- 2 Choose a date in the southwest monsoon - August is ideal, because the Somali Current is at its most energetic.
- 3 Keep the source on **satellite**. Point out that the eddies on screen are being found in real satellite currents, with no model reconstruction involved.
- 4 Open the eddy panel and look at 7.5 N, 53 E. A large anticyclone holds station there.
- 5 Switch the source to **glorys** and let the audience compare. Agreement between an independent observation product and a reanalysis is itself evidence.
- 6 Move to the fronts panel and read the printed threshold and mean gradient aloud - that pair is what stops a percentile detector from always claiming to have found something.

4. Why the problem statement needs this

The problem statement asks for a daily, eddy-resolving reconstruction that is useful over the Bay of Bengal and the Arabian Sea. Detecting the features themselves proves the daily product is being used for what daily data is uniquely good for - a monthly average erases every one of these.

It also gives the project a physical cross-check that does not depend on Argo. If the detectors find the basin's known circulation features in the right places at the right season, the underlying fields are behaving.

PS requirement	How this feature answers it
Daily temporal resolution	Only daily data can resolve features that move a few kilometres per day; this page is where that pays off.
Satellite-observation inputs	Eddies and fronts are computed from GLOBCURRENT currents and OSTIA SST - genuine observations.
Proof of concept over BoB and Arabian Sea	Both basins are covered, and their event regimes are visibly different.

5. The measured numbers behind it

What was found, with each claim carrying its own evidence tag - this feature is where the project is most careful about the difference:

Finding	Evidence tag	What that means
A ~243 km anticyclone holds station at 7.5 N, 53 E in August	[VERIFIED]	It is genuinely in the data and was measured here.
That eddy is the Great Whirl	[INFERRED]	It matches the standard description, but no paper was re-read on this machine, so we do not assert it as fact.
July's strongest gradient is 11.4 degC per 100 km at 11.4 N, 51.5 E	[VERIFIED] , and it sits in the Somali upwelling front region	Encouraging - and explicitly not evidence, because no front climatology was checked.
Upwelling is attributed to wind	Not claimed - the flag literally reads wind_attributed: False	The wind-stress product covers 2019-2022 and cannot reach our 2025-26 window.

- The fronts detector uses a percentile threshold, which by construction **always** returns the sharpest gradients present and can therefore never report 'no fronts'. Because that is a real weakness, the detector also returns the threshold and the mean gradient, so a flat field is visible as one.

6. The impact - and who it is for

Who	What changes for them
Fishing communities	Fronts and upwelling zones are where fish aggregate. Daily maps of them have direct, immediate livelihood value.
Cyclone forecasters	A warm-core eddy under a storm track is a known intensification hazard; a cold eddy is the opposite.
Navy and marine operations	Eddies bend sound and shift currents, affecting both acoustics and routing.
Ocean researchers	A daily, basin-wide, satellite-driven event census over a region where such products are scarce.

WHY SOCIETY GAINS FROM THIS

Small-scale fishers in India, Sri Lanka and the Gulf spend fuel searching for fish. Daily front and upwelling maps reduce that search, which cuts cost and risk for people going to sea in small boats. That is a direct, tangible benefit from a satellite product.

7. Say this to the jury

This finds the things that move - eddies, fronts, upwelling - from real satellite currents and real satellite temperature, daily. In August we find a 243-kilometre anticyclone holding station at 7.5 north, 53 east, exactly where the Somali Current's great eddy is described. Notice how we phrase that: the eddy is VERIFIED in the data, but calling it the Great Whirl is tagged INFERRED, because we have not re-read the paper on this machine. We tag every claim that way.

8. Questions the jury will ask

Q. Can you track an eddy over time, not just detect it?

A. Not yet, and we are precise about why. Until recently our data was monthly and tracking was genuinely impossible, because an eddy moves a few kilometres per day and consecutive months cannot be assumed to show the same feature. We now have 388 consecutive days, so tracking is **unbuilt rather than impossible**. We changed the wording when the data changed, rather than letting an old justification stand.

Q. Are your fronts validated?

A. No, and we say so. The detector runs and its strongest July gradient lands in a region where a front is expected, but no front climatology or published census was checked. That is encouraging, and it is not evidence.

Q. Does upwelling detection prove the wind caused it?

A. No. The wind-stress product we hold runs 2019 to 2022 and cannot reach our 2025-26 window, so the panel produces a signature and the attribution flag is literally set to false on screen.

9. The honest limit - say it before they find it

DETECTION ONLY, AND ONE DETECTOR CANNOT FAIL

Nothing on this page joins one date to the next, so there is no eddy tracking and no marine-heatwave product - both are unbuilt, not impossible. And the front detector's percentile threshold means it can never return an empty result, which is exactly why the threshold and the mean gradient are printed beside every map.

IN ONE LINE

Finds the moving features - eddies, fronts, upwelling - in genuine satellite fields, with every claim carrying its own evidence tag.

243 km

ANTICYCLONE AT 7.5 N, 53 E [VERIFIED]

11.4 degC

PER 100 KM, JULY'S STRONGEST FRONT

388 days

OF CONSECUTIVE DAILY FIELDS AVAILABLE

Feature 6 of 17. Next: Feature 7.